UNDER REVIEW

RESCIENCE C

Replication / Computational Biology

[Re] Evolutionary dynamics of cancer in response to targeted combination therapy

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Introduction

Bozic et al. [1] modelled the evolution of resistance in metastatic melanoma lesions treated with targeted therapy. A lesion grows as a multitype birth–death process from one sensitive cell; resistance to each drug arises by point mutation. From this model the authors derived closed-form probabilities that resistance is present when treatment starts and that a lesion is eradicated by one or two drugs (their equation 1), checked them against simulations, and drew clinical conclusions: a single mutation conferring cross-resistance to both drugs makes dual therapy fail in most large lesions, combinations must be given simultaneously rather than sequentially, and patients with a high tumour burden need three drugs. We re-implemented the model from the paper and its Materials and methods only (no code was published with the article) and compared our simulations with its analytic results (Figure 3), its patient-level predictions (Table 1), its simulations with a cost of resistance (Table 2) and its comparison of sequential and simultaneous therapy (Figure 5).

Methods

Model. As in [1]: cell types 00, 10, 01, 11 (1 = resistant to drug 1 or drug 2). Before treatment every cell divides at rate b and dies at rate d; the offspring of a 00 cell is of type 10, 01 or 11 with probabilities un_1, un_2, un_{12}, that of a 10 cell is 11 with probability $u(n_2 + n_{12})$, and symmetrically for 01. The lesion grows from one sensitive cell until it has M cells; treatment then starts, during which cells sensitive to at least one drug in use divide at b' and die at d', while fully resistant cells keep b and d. Parameters of Figure 3: $b = 0.14$, $d = 0.13$, $d' = 0.17$, $b' = b$ per day, $u = 10^{-9}$, $n_1 = n_2 = 50$, $n_{12} \in \{0, 1, 2\}$; for monotherapy, the types sensitive to drug 1 (00, 01) are treated.

Simulation. Like the authors, we use a hybrid scheme: a cell type with fewer than $K = 1000$ cells evolves stochastically, larger types deterministically, and mutants of the large types arise as Poisson events. Small types are advanced in steps of $\Delta t = 0.1$ day with Poisson numbers of births, deaths and mutant offspring (tau-leaping, $b\Delta t = 0.014$); an exact option simulates them event by event (Gillespie), as the authors did. A lesion whose founding lineage dies out before reaching M is restarted. After treatment starts, a lesion is eradicated when no cell is left and has failed when a resistant type reaches 10^5 cells. Each lesion has its own random stream (xoshiro256** [2], seeded from the run

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Code is available at https://github.com/quantumproteinsai/rescience-bozic2013-enderling2009 – DOI 10.5281/zenodo.23437598..

seed and the lesion index), so results do not depend on the number of threads. The code is C++ with OpenMP, driven from Python.

Extensions used by the paper. *Cost of resistance* (Table 2): a type with k resistance mutations has birth rate $(b-d)(1-c)^k + d$, with $d = b - 0.01$ and $d' = b + 0.03$. A cell made fully resistant by the single cross-resistance mutation carries one mutation and pays the cost once; one made resistant by two mutations pays it twice. *Sequential therapy* (Figure 5): drug 1 until relapse, defined as the lesion growing back to M cells after having shrunk below it, then drug 2. *Division-suppressing therapy*: $b' = 0.10$, $d' = 0.13$.

Comparison. Probabilities are compared with Wilson 95 % intervals [3]. The formulas of [1] were transcribed from the article's MathML. Patient lesion sizes come from its Supplementary file 2 (volume = long × short^2/2 cm^3, 10^9 cells/cm^3).

Results

Equation 1 (Figure 3). Equation 1 lies inside the 95 % interval of our simulation at 40 of 40 points (4,000 lesions each; largest absolute difference 0.009; Figure 1). The central result is reproduced: with one cross-resistance mutation, the probability of eradicating a lesion of 10^9 cells falls from 0.999 to 0.26.

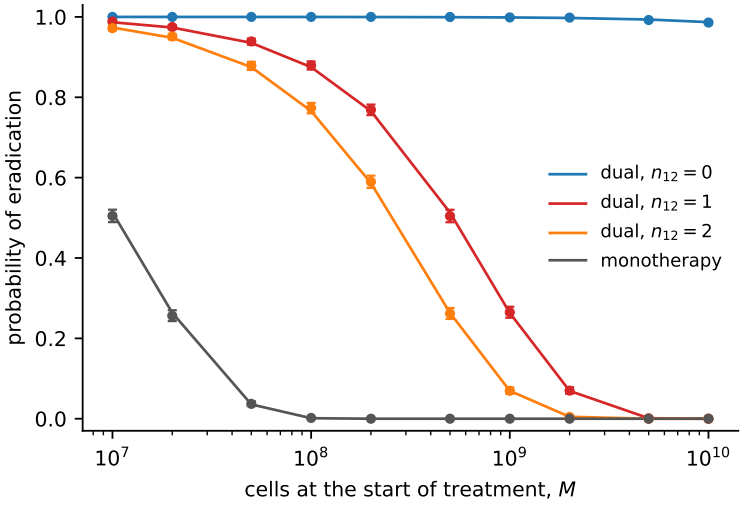


Figure 1. Probability of eradicating a lesion of M cells (parameters of Figure 3 of [1]). Points: simulation (4,000 lesions each, Wilson 95 % intervals); lines: equation (1).

Two-step resistance at detection. The probability that fully resistant cells are present when treatment starts is reproduced when resistance arises in one step ($n_{12} \geq 1$), but not when it needs two mutations ($n_{12} = 0$, $M = 10^9$): simulation 0.0029 [0.0022, 0.0037], formula 0.0036 (Table 1). The difference remains with a five times smaller time step, a ten times higher stochastic threshold, and exact event-by-event simulation, so it is not an artefact of our scheme; the corresponding formula is an approximation in [1]. The probability of eradication, which depends on surviving resistant lineages rather than on cells present at one instant, agrees.

Table 1. Two-step resistance present at the start of treatment ($n_{12} = 0$, $M = 10^9$, 20,000 lesions each).

small populations	Δt (day)	threshold K	simulated (95 % CI)	formula $1 - p_1 p_2$
tau-leaping	0.1	1,000	0.0029 [0.0022, 0.0037]	0.0036
tau-leaping	0.1	10,000	0.0033 [0.0026, 0.0042]	0.0036
tau-leaping	0.02	1,000	0.0022 [0.0016, 0.0030]	0.0036
exact events	0.1	1,000	0.0023 [0.0017, 0.0031]	0.0036

Patients (Table 1). Recomputing the failure probability of dual therapy from the lesion sizes of Supplementary file 2 reproduces Table 1 within 0.0015 for 21 of 22 patients (Table 2); simulating every lesion (1,000 replicates per patient) agrees with equation (1) in 42 of 44 comparisons. For patient N20, the three lesions of the supplementary file total 1.29×10^9 cells, giving 0.821 with one cross-resistance mutation, whereas Table 1 lists 1.6×10^9 cells and 0.881, which is what equation (1) gives for that larger burden. The paper's two sources disagree for this patient; its conclusions are unaffected.

Table 2. Probability that dual therapy fails, per patient: Table 1 of [1], equation (1) with the lesions of Supplementary file 2, and simulation.

			dual, $n_{12} = 1$			dual, $n_{12} = 0$		
patient	lesions	cells	Table 1	eq. (1)	simulated	Table 1	eq. (1)	simulated
N1	18	2.6×10^{11}	1.000	1.000	1.000	0.283	0.283	0.258
N2	25	2.3×10^{11}	1.000	1.000	1.000	0.260	0.260	0.256
N3	26	1.7×10^{11}	1.000	1.000	1.000	0.203	0.203	0.182
N4	30	1.4×10^{11}	1.000	1.000	1.000	0.172	0.172	0.175
N5	21	1×10^{11}	1.000	1.000	1.000	0.128	0.128	0.119
N6	8	9.8×10^{10}	1.000	1.000	1.000	0.120	0.120	0.118
N7	25	9.1×10^{10}	1.000	1.000	1.000	0.112	0.112	0.097
N8	8	7.4×10^{10}	1.000	1.000	1.000	0.092	0.091	0.093
N9	23	6.4×10^{10}	1.000	1.000	1.000	0.080	0.080	0.071
N10	5	5.5×10^{10}	1.000	1.000	1.000	0.069	0.069	0.079
N11	14	5.4×10^{10}	1.000	1.000	1.000	0.068	0.068	0.058
N12	23	4.8×10^{10}	1.000	1.000	1.000	0.061	0.061	0.071
N13	9	4.1×10^{10}	1.000	1.000	1.000	0.052	0.052	0.055
N14	13	4.1×10^{10}	1.000	1.000	1.000	0.051	0.051	0.047
N15	8	3.4×10^{10}	1.000	1.000	1.000	0.042	0.043	0.027
N16	7	2.2×10^{10}	1.000	1.000	1.000	0.028	0.028	0.029
N17	10	2.1×10^{10}	1.000	1.000	1.000	0.027	0.027	0.024
N18	4	2×10^{10}	1.000	1.000	1.000	0.026	0.026	0.018
N19	9	1.8×10^{10}	1.000	1.000	1.000	0.023	0.023	0.018
N20	3	1.3×10^{9}	0.881	0.821	0.825	0.002	0.002	0.002
N21	21	1.3×10^{9}	0.828	0.828	0.811	0.002	0.002	0.001
N22	1	8.5×10^{8}	0.677	0.677	0.709	0.001	0.001	0.001

Cost of resistance (Table 2). All 32 of 32 entries agree with the paper, whose values are given to two decimals (10,000 lesions per entry, against 10^6 in the paper; Table 3). Counting the cost per mutation matters: when every fully resistant cell paid it twice, the failure probabilities for $n_{12} = 1$, $M = 10^9$ and $c = 5$ and 10 % were 0.70 and 0.66 instead of the paper's 0.72 and 0.70.

Sequential versus simultaneous therapy (Figure 5). With one cross-resistance mutation, simultaneous therapy eradicates 26.3 % of lesions of 10^9 cells and sequential therapy none; without cross-resistance, simultaneous therapy eradicates almost all lesions and sequential therapy none, as reported (Table 4). A therapy suppressing division instead of killing cells eradicates 28.9 % (paper: 29 %).

Table 3. Probability of treatment failure with a cost of resistance c per mutation: simulation (Table 2 of [1] in parentheses).

n_{12}	M	b	$c = 0\,\%$	$c = 1\,\%$	$c = 5\,\%$	$c = 10\,\%$
0	10^9	0.14	0.001 (0.00)	0.001 (0.00)	0.001 (0.00)	0.001 (0.00)
0	10^9	1	0.006 (0.01)	0.007 (0.01)	0.006 (0.01)	0.003 (0.00)
1	10^9	0.14	0.737 (0.74)	0.739 (0.73)	0.720 (0.72)	0.695 (0.70)
1	10^9	1	0.739 (0.74)	0.735 (0.74)	0.720 (0.72)	0.697 (0.70)
0	10^{11}	0.14	0.117 (0.12)	0.107 (0.11)	0.080 (0.08)	0.054 (0.06)
0	10^{11}	1	0.533 (0.53)	0.509 (0.51)	0.420 (0.42)	0.319 (0.32)
1	10^{11}	0.14	1.000 (1.00)	1.000 (1.00)	1.000 (1.00)	1.000 (1.00)
1	10^{11}	1	1.000 (1.00)	1.000 (1.00)	1.000 (1.00)	1.000 (1.00)

Table 4. Sequential and simultaneous dual therapy, $M = 10^9$ (10,000 lesions each).

Therapy	n_{12}	eradicated (95 % CI)	paper
simultaneous	1	0.263 [0.254, 0.271]	26.3 % eradicated
sequential	1	0.000 [0.000, 0.0004]	0 % (73.7 % 11 at start)
simultaneous	0	0.999 [0.998, 0.999]	$>$ 99 % eradicated
sequential	0	0.000 [0.000, 0.0004]	0 %
division suppressed ($b' = 0.10$, $d' = 0.13$)	1	0.289 [0.280, 0.298]	29 % (26 % by killing)

Discussion

The model of [1] could be re-implemented from its description, and its main results and clinical conclusions are reproduced quantitatively: equation 1, the patient-level predictions, the effect of a cost of resistance, and the failure of sequential therapy. Two points needed care or could not be reproduced. First, the cost of resistance must be counted per mutation, which the text implies but does not state for cross-resistant cells. Second, the approximation for two-step resistance present at detection overestimates it by 8–39 %; it should not be used to check a simulation, whereas equation 1 can be. The disagreement between Table 1 and Supplementary file 2 for one patient does not change any conclusion. We contacted the corresponding author about these points.

Code, data and reproducibility

Code: https://github.com/quantumproteinsai/rescience-bozic2013-enderling2009, archived at doi:10.5281/zenodo.23437598. The command `make all` rebuilds the programs and regenerates every number, table and figure from fixed seeds; a fresh clone reproduces all data files identically. For reviewers, `make quick-check` reproduces 11 representative published results exactly, from scratch, in about two minutes on a laptop. The lesion sizes of Supplementary file 2 of [1] are redistributed under its CC BY licence.

Use of AI assistance

The software, simulations, analyses and the first draft of this article were produced with an AI coding assistant (Claude Opus 5.5, Anthropic, used through Claude Code), under the direction of the author, who defined the study, chose the comparisons, checked the results against the original publication and revised the text. The AI is not an author; the author takes full responsibility for the content.

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